A Quick Synopsis of the Letter Spirit Project: High-Level Perception and Creation of Letterforms Using Fluid Concepts

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The Letter Spirit project is an attempt to model central aspects of human high-level perception[1] and creativity on a computer. We believe creativity is an automatic consequence of the existence of sufficiently flexible and context-sensitive concepts. Accordingly, our goal is to implement a model of *fluid concepts*. Such a model will, we believe, shed light on human creativity.

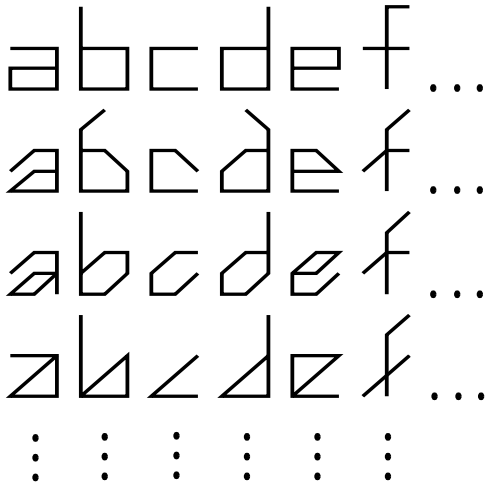


Figure 1: Items in any column have *letter* in common. Items in any row have *spirit* in common.

Letter Spirit explores the creative act of artistic letter-design. The aim is to model how the 26 lowercase letters of the roman alphabet can be rendered in many different but internally coherent styles. The program addresses two important aspects of letterforms: the *categorical sameness* possessed by letters belonging to a given category (e.g., 'a') and the *stylistic sameness* possessed by letters belonging to a given style (e.g., Helvetica). Figure 1 illustrates these two ideas and shows their relationship. The program will start with one or more seed letters representing a style, and create the rest of the letters in such a way that all 26 share the same style, or *spirit*.

To avoid the need for modeling low-level vision and to focus attention on the deep aspects of letter design, we eliminated all continuous variables, leaving only a small number of deep and discrete decisions affecting each letterform. Specifically, letterforms are restricted to short line

[1]High-level perception is the perceptual process whereby sensory data activate concepts at various levels of abstraction. For a discussion of high-level perception see [Chalmers, French, and Hofstadter, 1990].

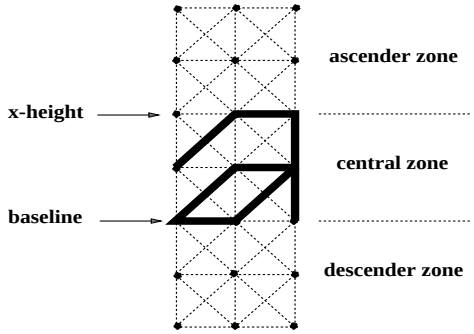


Figure 2: The grid, with one of the many possible sets of quanta instantiating an 'a' turned on.

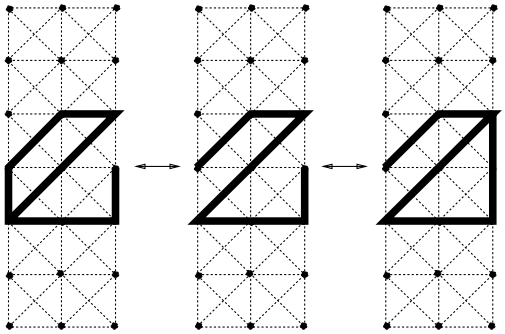


Figure 3: Rapid metamorphosis of an 'e' into an 'a' via 'z', with merely a one-quantum change in each step.

segments on a fixed grid having 21 points arranged in a 3×7 array. Legal line segments, called *quanta*, are those that connect any point to any of its nearest neighbors horizontally, vertically, or diagonally. There are 56 possible quanta on the grid, as shown in Figure 2.

Since any quantum is either on or off, decisions on the grid are coarse. Surprisingly, the variety among letters of a given category is still huge — hundreds of versions of each letter and around 600 complete gridfonts have been designed by humans.[2] Almost paradoxically, the Letter Spirit domain's severe limitations seem to engender this diversity.

It is impossible to make tiny changes, so any two instantiations of a given letter are significantly different from each other. Indeed, because of the coarseness of the grid, designers are forced to tamper with the conceptual essence of letters. Decisions to add or subtract even one quantum often fundamentally affect category membership. Figure 3 illustrates the large distance in letter-space one can travel with minute changes on the grid. A "small" change in the letterform — the erasure of just one quantum — changes the shape from being a strong 'e' to a strong 'z'. A similar one-quantum addition transforms the 'z' into an 'a'. So category membership in the Letter Spirit domain is a tricky matter. In fact, modeling the process of deciding which of the 26 categories a given shape on the grid belongs to (if any) is one of the hardest aspects of the project.

Because there are no surface features to manipulate at a fine-grained level, one ends up playing at the boundaries of the 26 categories. Consequently, many gridfonts are wild, sometimes having angular, blocky, spiky, sparse, or otherwise bizarre letters. We are not striving for typefaces with high readability or letterforms beautiful at the surface level. Rather, we are attempting to understand the conceptual nature of letterforms and thereby gain insight into what imbues concepts *in general* with their fluidity. Pushing the 26 categories to their edges often results in an intellectual beauty not localized in any *single* letterform of a gridfont, but spread out over the gridfont *as a whole* — the beauty of *spirit* rather than of *letter*.

We use the term *letter-conceptualization* to denote the abstract idea for drawing a letter. A letter-conceptualization is present in a designer's mind before any marks are put on paper. The actual shape drawn on paper is a *letterform*. Letter Spirit is concerned with playing with letter-conceptualizations and instantiating them as letterforms.

[2]Two representative human-designed gridfonts are included at the end of this abstract.

To make the shape/conceptualization distinction more vivid, consider lowercase 'x'. For most US-educated adults, the conceptualization of 'x' consists of a forward slash and a backward slash of the same size that cross somewhere near the middle. What is in people's minds is not a *picture* of two lines, but a set of *ideas*. By contrast, English schoolchildren are taught to draw a lowercase cursive 'x' as two small crescents facing away from each other but "kissing" in the middle. If we look at an 'x' in this way, we are suddenly struck by this new conceptualization. The shape drawn on paper is the same, but what is in our mind is different. This brings up the question: Are some conceptualizations of letters better than others? Is one of them the *right* way to think about 'x'?

Although there is no "right" conceptualization for a given letter, some conceptualizations are better than others. A conceptualization's quality depends on how strongly the spectrum of instances it defines are rated as members of the intended category. The conceptualizations defining the strongest instances are more important than their rivals, and provide *norms* for a given letter.

The conceptual pieces into which a letter is broken in the mind's eye are called *roles*. For example, the two crossing slashes in an imagined 'x' are roles. So also are their four tips, and the crossing-point in the middle. Each role has a different degree of *criteriality* to the letter — the degree to which its presence or absence matters. Of course, roles are not realized in a black-and-white manner any more than entire letters are. Different shapes instantiate a given role more strongly or more weakly than others.

The internal structure of a category is represented as a collection of interacting roles. Category membership at the letter level is partially determined by category membership at the lower level of roles. In addition, *stylistic* appropriateness of a shape is judged in terms of *how* roles inside letters are filled — ways in which *norms are violated*. For instance, consider the role *crossbar*, which belongs to conceptualizations of 'e', 'f', and 't' (at least). Norm violations include the following: "crossbar too high"; "crossbar tilted upwards"; "unusually short crossbar"; "crossbar missing"; etc. Any such violation is a stylistic hallmark that must be respected and propagated (via analogy) to other letters, even ones that lack the crossbar role.

For a person, designing a gridfont takes between ten minutes and three hours (after which the font remains potentially subject to scrutiny and minor revision). The process involves myriad small operations, ranging from the mechanical to the inventive. A typical action (e.g., creating a 'k' or even changing a single quantum in an already-designed letter) sets off repercussions that echo throughout the continuing design process, all over the gridfont, and at all levels of abstraction. This activity is largely guided by *a priori* notions of the interrelatedness of letter categories (e.g., 'd' and 'b' are often considered near-reflections of each other). Many such actions occur, and eventually a stable gridfont emerges.

Dissatisfaction and contradiction either within a given letterform or between letters are the prime movers behind the design process. If two or more letters seem to conflict stylistically, the resulting gridfont will be unsatisfying to a human. Letter Spirit must thus be capable of *recognizing* and *resolving* such conflicts, in order to create a coherent style. Sometimes the conflicts are subtle, and therefore require refined artistic judgment-calls. Modeling the ability to find conflicts, diagnose them, and convert diagnoses of problems into reasonable suggestions for solutions is a key aspect of the project.

Any design decision will affect not only the letter currently under consideration but also conceptually close letters. For example, a design decision regarding the "post" of a 'b' is likely to have enormous effect on the post of the 'd', as well as noticeable influence on the descenders of the 'p' and the 'q'. Of course, the extent and type of this influence are highly dependent on the

particular style and are in no way mechanical or formulaic. Most aspects of letterforms are likely to propagate their influence to varying extents through the entire gridfont. The propagating wave will probably cause many retroactive adjustments (both major and minor) to some "already finished" letters, and give rise to ideas for letters not yet designed. One design decision will typically spark others, which in turn will engender others, and so on.

Eventually, when enough "decision waves" have washed over the entire gridfont, letterforms begin to have a high degree of internal consistency, and a style begins to emerge. Once the tension of inconsistency eases up, no more large-scale changes are required. Minor adjustments may continue, but for the most part, the large-scale creative act will be finished. This temporally-extended, serial aspect of creation is an indispensable part of true creativity.

Letter Spirit will have two major modules: the *recognizer–evaluator* and the *juggler–suggestor*. The recognizer–evaluator, much of which has been implemented in a rudimentary way, is intended to model letter perception at a very sophisticated level. The juggler–suggestor will play with letter-conceptualizations and suggest new letterforms for inclusion in the current gridfont.

Recognition in the recognizer–evaluator must go far beyond that of OCR programs, which deal with fairly bland letters relative to those in typical gridfonts. (By contrast, gridfont letters play at the very edges of recognizability even for humans.) The recognizer–evaluator must be able to evaluate a letterform both in terms of its fit within a given letter *category* and in terms of its fit within a particular *style*. It must decide whether the balance between letter and spirit in a proposed letterform is acceptable. Each letterform will receive an evaluation summarizing the program's happiness with the letter.

At the outset of a typical Letter Spirit run, one or more seed letters will be provided to the recognizer–evaluator, which must decide which letter category the shape belongs to, judge *how good an instance* of that category the letterform is, and make note of *stylistic tendencies* in the letterform. Stylistic tendencies will then be made available to the juggler–suggestor, which will suggest one or more new letters to add to the current gridfont. Such new letters will engender further stylistic reverberations among the already-existing letters of the emerging gridfont, perhaps changing existing letters' evaluations, thus forcing a new round of adjustments.

The juggler–suggestor's responsibility is to take *incoming* norm violations and produce suggestions for *outgoing* norm violations. This can be a simple or sophisticated process, depending on how close the two letterforms are. For instance, a short crossbar on a 't' would easily inspire a short crossbar on the 'f', but it is harder to export this norm violation to letters such as 'o' or 'k'. However, with a variety of conceptualizations available, one of them might lend itself to a direct analogy even if the others do not. Thus it is crucial that the juggler–suggestor recognize the appropriateness of shifting from a default conceptualization to a more unusual one, under pressure from observed norm violations. Furthermore — and this is where the highest degree of creativity in a human designer is called for — the juggler–suggestor should not be limited to a fixed set of *a priori* conceptualizations for each letter. It should be able to pull apart and reassemble roles and their relationships, thus coming up with unanticipated conceptualizations (such as seeing 'x' as a 'v' perched on top of an inverted 'v'). *Manipulation of roles* is at the heart of our model of the creative process.

The juggler–suggestor will make its suggestions under stylistic constraints discovered by the recognizer–evaluator. Shapes it proposes will in turn be evaluated by the recognizer–evaluator. Letters that pass inspection will be inserted into the gridfont, while letters rejected by the recognizer–evaluator will need to be modified by the juggler–suggestor. Any such modification

will have to be based on failure information provided by the recognizer–evaluator. The recognizer–evaluator and the juggler–suggestor will work in tandem during the creation of a gridfont, giving rise to a model of the design process involving the interaction of perception and suggestion.

Since a design decision on the letter 'e' (for instance) has potential implications for all 25 other letters, one could decide arbitrarily whether to work on 'a' or 'z', etc. next — and in fact, the program will choose in a biased probabilistic fashion which letter to tackle next. Instead of there being a central focus of control, multiple paths will to some extent be explored in parallel (with the most fruitful paths being allowed to advance most rapidly). This is especially true of within-letter processing.

The two main predecessors of Letter Spirit are Copycat [Mitchell, 1990] and Tabletop [French, 1992] both of which create microdomain analogies in psychologically plausible ways. Both are based on models of high-level perception making use of a novel type of architecture that falls between the symbolic and connectionist paradigms. Top-level behavior emerges from many low-level stochastic computational actions that occur in parallel. Letter Spirit will make use of a similar architecture, but will deal much more with the internal structure of concepts, and will focus on a larger-scale creative process that unfolds over an extended period of time.

Letter Spirit will test the applicability of the Slipnet/Coderack/Workspace architecture to large-scale creative tasks. The parallel stochastic processing mechanisms of Letter Spirit fall under the rubric of *emergent computation*, wherein complex high-level behavior emerges as a statistical consequence of many small computational actions. Like Copycat and Tabletop, Letter Spirit will occupy a level of cognitive modeling somewhere between connectionism and symbolic AI — the level we feel is the most useful for the understanding of high-level perception, the fluidity of concepts, and creativity.

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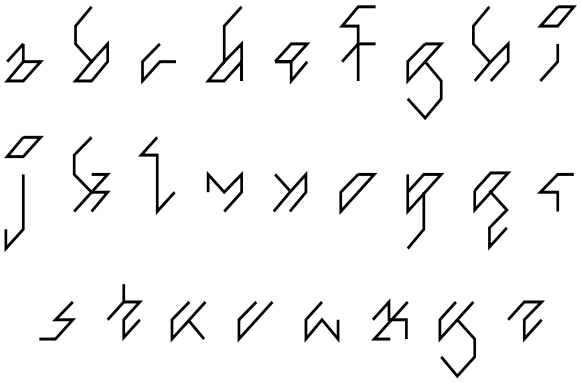
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Swallowtail



Intersect

